

Human Motor Cortex Gene Expression/h1>

AUTHOR

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Introduction and Motivation

The study of gene expression patterns in the neocortex is a vital and very interesting area of neuroscience research because it has the potential to reveal how different neuronal subpopulations are organized and function within the human brain, thereby contributing to our overall understanding of the brain.

This dataset comes from a study about single-nucleus transcriptomes from the human primary motor cortex (M1) and contains 76,533 nuclei from two post-mortem human samples. This project utilizes the files that contain the trimmed means by cluster and metadata. The high-resolution transcriptomic data obtained in this study is going to allow us to identify distinct neuronal subpopulations based on their gene expression profiles thereby offering insights into the cellular diversity within the M1 cortex.

The motivation for examining the M1 cortex lies in its critical role in voluntary movement and motor control, as well as its involvement in disorders such as amyotrophic lateral sclerosis (ALS) and Parkinson's disease. More than that, studying the molecular diversity in this region of the brain aligns with my desire to understand the molecular basis of brain function.

Literature Context

Previous research has shown that neuronal populations in the neocortex can be differentiated based on their gene expression profiles and laminar distribution. Watakabe et al. (2007) examined layer-specific gene expression in the mammalian neocortex where it compared between monkeys and mice. This study identified genes like ER81, Nurr1, CTGF, and 5-HT2C, which it found to exhibit both conserved layer preferences and significant interspecies differences. Essentially, these findings highlighted how specific neuronal subpopulations can be categorized based on gene expression, offering insights into understanding evolutionary adaptations.

Miller et al. (2019), on the other hand, discussed how the cellular diversity of the human cerebral cortex emerges through development and is organized in a way that supports complex cognitive functions. Their findings support the idea that cortical organization can vary greatly among species, while maintaining fundamental features that are conserved.

Vawter et al. (2004) focused on gender-specific gene expression in post-mortem human brains, highlighting how differential expression of X- and Y-linked genes can be detected using microarray technology. They found that certain genes linked to sex chromosomes show distinct patterns of expression between males and females, suggesting that sex-specific gene expression might play a role in neural function and potential vulnerability to neuropsychiatric disorders.

Jacques et al. (2011) investigated dynamic neural networks that supported memory retrieval and found that specific cortical regions do indeed show unique gene expression profiles that correspond with their

functions. This study lends great support to my approach to examining the M1 cortex, with aims to map the molecular architecture that correlates with motor function and its distinct characteristics compared to other cortical regions.

As such, these are the four hypothesis, my project focuses on

1. Gene expression profiles differ significantly between SST and VIP neuron subclasses.
2. Donor sex may result in small but noticeable differences in gene expression within some neuron types.
3. Genes involved in neurotransmission will vary across different cortical layers.
4. Specific marker genes are strongly associated with GABAergic, Glutamatergic, or Non-neuronal cell classes.

By using the combined insights from these earlier studies and the high-resolution transcriptomic data, I aim to understand more about the M1 regions by seeing if the data analysis done in this project can provide some insight into the chose hypothesis there furthering my knowledge of the M1 cortex.

Methods and Data Wrangling.

The statistical analysis followed a structured approach in order to ensure consistency and accuracy across all hypotheses. It began by importing the two data sets by them read into R studio followed by an examinations of these datasets using `glimpse()` and `summary()` functions to get an overview of the data structure.

For each hypothesis, the relevant data subset was extracted. For example in H1, SST and VIP neurons were extracted, for H2, I looked specifically at the male and female samples. For H3, the focus was on neurotransmission-related genes across various cortical layers, and for H4, I only looked at gene expression data by GABAergic, Glutamatergic, and Non-neuronal cell classes.

After the relevant columns were selected, the data was cleaned. I calculated the proportion of zeros per gene, then filtered out genes with more than 95% zero values. This was all done to reduce noise so that the statistical analysis was focused only on genes that provided meaningful expression levels.

Once the data was cleaned, depending on the number of groups that we were comparing, the relevant statistical tests for each hypothesis was performed. Only two types of tests have been used within this project. One of them being the Two-Sample T-Tests for comparing two groups (like male vs. female) and ANOVA for comparing multiple groups (like different cortical layers or cell classes). After obtaining p-values, only the results that showed a significance ($p < 0.05$) were kept.

Finally, I also visualized the significant results using various plots. For H1 and H2, double bar plots were used to compare expression levels between groups. For H3, a boxplot was used to visualize gene expression across cortical layers. And finally For H4, a boxplot as well as a density plot to show gene expression variability among cell classes.

Results and Analysis

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2     3.5.1    ✓ tibble     3.2.1
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr       1.0.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(viridis)
```

Loading required package: viridisLite

```
library(dplyr)
library(ggplot2)

trimmed_means_data <- read_csv("C:\\Users\\Alex Nite\\Desktop\\Spring 2025\\Data Science\\trimmed_
```

Rows: 50281 Columns: 128

```
— Column specification —
Delimiter: ","
chr  (1): feature
dbl (127): Inh L1 PAX6 CHRFAM7A, Inh L1-3 VIP CBLN1, Exc L3 LAMP5 CARM1P1, I...
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
metadata <- read_csv("C:\\Users\\Alex Nite\\Desktop\\Spring 2025\\Data Science\\metadata.csv")
```

Rows: 76533 Columns: 39

```
— Column specification —
Delimiter: ","
chr (25): sample_name, exp_component_name, cluster_label, cluster_color, cla...
dbl (11): cluster_order, class_order, subclass_order, donor_sex_order, regio...
lgl  (3): full_genotype_label, outlier_call, outlier_type
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Looking at the dataset
```

```
glimpse(metadata)
```

```
Rows: 76,533
```

```
Columns: 39
```

```
$ sample_name      <chr> "AAACCCAAGGATTTCC-LKTX_190129_01_A01", "AA...
$ exp_component_name <chr> "AAACCCAAGGATTTCC-21L8TX_180927_001_A01", ...
$ cluster_label    <chr> "Inh L1-2 SST CCNJL", "Exc L5-6 FEZF2 IFNG...
$ cluster_color    <chr> "#fb8d00", "#2c815f", "#547d7a", "#cecd32"...
$ cluster_order    <dbl> 50, 116, 87, 75, 120, 101, 111, 90, 90, 10...
$ class_label      <chr> "GABAergic", "Glutamatergic", "Glutamaterg...
$ class_color      <chr> "#FF7373", "#3DCC3D", "#3DCC3D", "#3DCC3D"...
$ class_order      <dbl> 1, 2, 2, 2, 3, 2, 2, 2, 2, 2, 2, 2, 1, 2, ...
$ subclass_label   <chr> "Sst", "L5/6 NP", "L5 IT", "L2/3 IT", "Oli...
$ subclass_color   <chr> "#FF9900", "#3E9E64", "#50B2AD", "#C4EC04"...
$ subclass_order   <dbl> 5, 12, 8, 7, 17, 13, 11, 8, 8, 13, 7, 7, 5...
$ donor_sex_label  <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F...
$ donor_sex_color  <chr> "#FF7373", "#FF7373", "#FF7373", "#FF7373"...
$ donor_sex_order  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
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$ region_color     <chr> "#FF7373", "#FF7373", "#FF7373", "#FF7373"...
$ region_order     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
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$ cortical_layer_color <chr> "#FF7373", "#FF7373", "#FF7373", "#FF7373"...
$ cortical_layer_order <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ cell_type_accession_label <chr> "CS1912131050", "CS1912131116", "CS1912131...
$ cell_type_accession_color <chr> "#fb8d00", "#2c815f", "#547d7a", "#cecd32"...
$ cell_type_accession_order <dbl> 50, 116, 87, 75, 120, 101, 111, 90, 90, 10...
$ cell_type_alias_label <chr> "Inh L1-2 SST CCNJL", "Exc L5-6 FEZF2 IFNG...
$ cell_type_alias_color <chr> "#fb8d00", "#2c815f", "#547d7a", "#cecd32"...
$ cell_type_alias_order <dbl> 50, 116, 87, 75, 120, 101, 111, 90, 90, 10...
$ cell_type_alt_alias_label <chr> NA, NA, NA, NA, NA, "L6 CT_2", NA, NA, NA,...
$ cell_type_alt_alias_color <chr> "#fb8d00", "#2c815f", "#547d7a", "#cecd32"...
$ cell_type_alt_alias_order <dbl> 50, 116, 87, 75, 120, 101, 111, 90, 90, 10...
$ cell_type_designation_label <chr> "Neuron 50", "Neuron 116", "Neuron 87", "N...
$ cell_type_designation_color <chr> "#fb8d00", "#2c815f", "#547d7a", "#cecd32"...
$ cell_type_designation_order <dbl> 50, 116, 87, 75, 120, 101, 111, 90, 90, 10...
$ external_donor_name_label <chr> "H18.30.001", "H18.30.001", "H18.30.001", ...
$ external_donor_name_color <chr> "#FF7373", "#FF7373", "#FF7373", "#FF7373"...
$ external_donor_name_order <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ specimen_type    <chr> "nucleus", "nucleus", "nucleus", "nucleus"...
$ full_genotype_label <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
$ outlier_call     <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, ...
$ outlier_type     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
```

```
glimpse(trimmed_means_data)
```

Rows: 50,281

Columns: 128

\$ feature	<chr> "DDX11L1", "WASH7P", "MIR6859-1", "MIR13...
\$ `Inh L1 PAX6 CHRFA7A`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-3 VIP CBLN1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3 LAMP5 CARM1P1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-3 SST FAM20A`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-6 LAMP5 AARD`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-6 LAMP5 CA1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L6 THEMIS SLN`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L2 VIP SLC6A16`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5 FEZF2 NREP-AS1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L3-5 VIP IGDCC3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 SST FBN2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L3-5 VIP HS3ST3A1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-6 PVALB COL15A1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L2-3 LINC00507 DSG3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3-5 RORB TNNT2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L6 FEZF2 FFAR4`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L2 LINC00507 ATP7B`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L3-6 VIP ZIM2-AS1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 PVALB GAPDHP60`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5-6 THEMIS SMYD1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L2-3 RORB PTPN3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `VLMC L1-5 PDGFRA COLEC12`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5 FEZF2 RNF144A-AS1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Inh L3-6 VIP UG0898H09`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Exc L3-5 RORB LAMA4`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3-5 RORB LNX2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Micro L1-6 TYROBP CD74`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5-6 FEZF2 FILIP1L`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 LAMP5 NMBR`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L6 FEZF2 KLK7`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5 FEZF2 PKD2L1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L2-5 VIP SOX11`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5 FEZF2 CSN1S1`	<dbl> 0.00000000, 0.00000000, 0.00000000, 0.00...
\$ `Inh L3-5 SST CDH3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L2 PVALB FRZB`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 PVALB SST CRHR2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-2 VIP SCML4`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Astro L1-6 FGFR3 AQP1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L2 LAMP5 KCNG3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 LAMP5 BMP2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Inh L3-5 SST GGTLC3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L3-5 VIP TAC3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 SST ISX`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5-6 THEMIS TNFAIP6`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Oligo L2-6 OPALIN FTH1P3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3-5 RORB RPRM`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5-6 FEZF2 C9orf135-AS1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-3 VIP CHRNA2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3-5 RORB LINC01202`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Exc L5-6 FEZF2 LPO`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 LAMP5 PVRL2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Astro L1-6 FGFR3 PLCG1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L2-5 PVALB HHIPL1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 VIP COL4A3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Oligo L5-6 OPALIN LDLRAP1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 PVALB KCNIP2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L5 THEMIS LINC01116`	<dbl> 0.00000000, 0.00000000, 0.00000000, 0.000000...
\$ `Inh L5-6 SST BEAN1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Exc L3-5 FEZF2 ASGR2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Inh L1 PAX6 MIR101-1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Inh L1-2 VIP PTGER3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L3-6 PAX6 LINC01497`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Oligo L2-6 OPALIN MAP6D1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 LAMP5 RAB11FIP1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L6 THEMIS SNTG2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L3 RORB OTOGL`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Exc L6 FEZF2 PDYN`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `OPC L1-6 PDGFRA COL20A1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 PVALB ZFPM2-AS1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 VIP KLHDC8B`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-6 VIP SLC7A6OS`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 SST DEFB108B`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 SST PAWR`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L2 PAX6 FREM2`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
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\$ `Inh L1-2 PVALB CDK20`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-2 VIP WNT4`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L5-6 LAMP5 CRABP1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1 SST P4HA3`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
\$ `Inh L1-3 VIP FNDC1`	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...

```

$ `Inh L1-6 SST NPY` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Oligo L3-6 OPALIN ENPP6` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L1-2 SST CCNJL` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L1-5 VIP SMOC1` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5 THEMIS SLC22A18` <dbl> 0.00000000, 0.00000000, 0.00000000, 0.00...
$ `Inh L1-5 VIP LINC01013` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L2-5 VIP BSPRY` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L5-6 PVALB MEPE` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L2-3 RORB CCDC68` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L5 SST RPL35AP11` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Endo L2-5 NOSTRIN SRGN` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L2-3 SST NMU` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L6 THEMIS LINC00343` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5 THEMIS FGF10` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Astro L1 FGFR3 SERPINI2` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L1-2 VIP EXPH5` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L1-5 VIP PHLDB3` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L5-6 SST KLHL1` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5-6 FEZF2 CFTR` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5-6 FEZF2 SH2D1B` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L2-5 PVALB RPH3AL` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L3 THEMIS ENPEP` <dbl> 0.00000000, 0.00000000, 0.00000000, 0.00...
$ `Exc L6 FEZF2 PROKR2` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L1-2 VIP HTR3A` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5-6 FEZF2 OR1L8` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L2 LINC00507 GLRA3` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Inh L3-5 SST OR5AH1P` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L2-3 RORB RTKN2` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ `Exc L5 RORB MED8` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...

```

```
summary(trimmed_means_data)
```

feature	Inh L1 PAX6 CHRFAM7A	Inh L1-3 VIP CBLN1
Length:50281	Min. : 0.0000	Min. : 0.0000
Class :character	1st Qu.: 0.0000	1st Qu.: 0.0000
Mode :character	Median : 0.0000	Median : 0.0000
	Mean : 0.2749	Mean : 0.3339
	3rd Qu.: 0.0000	3rd Qu.: 0.0000
	Max. :14.8999	Max. :14.8129

Exc L3 LAMP5 CARM1P1	Inh L1-3 SST FAM20A	Inh L1-6 LAMP5 AARD
Min. : 0.0000	Min. : 0.000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.000	Median : 0.0000
Mean : 0.6069	Mean : 0.226	Mean : 0.4289
3rd Qu.: 0.0000	3rd Qu.: 0.000	3rd Qu.: 0.0000
Max. :15.2007	Max. :14.690	Max. :15.0136

Inh L1-6 LAMP5 CA1	Exc L6 THEMIS SLN	Inh L2 VIP SLC6A16	Exc L5 FEZF2 NREP-AS1
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.0000

Mean : 0.4157	Mean : 0.5879	Mean : 0.4001	Mean : 0.5921
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.9595	Max. :15.5100	Max. :14.3410	Max. :15.3849
Inh L3-5 VIP IGDCC3 Inh L5-6 SST FBN2 Inh L3-5 VIP HS3ST3A1			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.3275	Mean : 0.3353	Mean : 0.3426	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :14.6805	Max. :15.4086	Max. :14.9849	
Inh L1-6 PVALB COL15A1 Exc L2-3 LINC00507 DSG3 Exc L3-5 RORB TNNT2			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.3763	Mean : 0.5922	Mean : 0.5844	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.4566	Max. :14.9933	Max. :15.2085	
Exc L6 FEZF2 FFAR4 Exc L2 LINC00507 ATP7B Inh L3-6 VIP ZIM2-AS1			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.5598	Mean : 0.4869	Mean : 0.4484	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :14.7196	Max. :15.0252	Max. :14.9234	
Inh L5-6 PVALB GAPDHP60 Exc L5-6 THEMIS SMYD1 Exc L2-3 RORB PTPN3			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.4428	Mean : 0.5976	Mean : 0.4465	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.0244	Max. :15.0928	Max. :15.1382	
VLMC L1-5 PDGFRA COLEC12 Exc L5 FEZF2 RNF144A-AS1 Inh L6 SST TH			
Min. : 0.00000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.00000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.00000	Median : 0.0000	Median : 0.0000	
Mean : 0.06765	Mean : 0.5064	Mean : 0.5075	
3rd Qu.: 0.00000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.41100	Max. :15.3796	Max. :14.7971	
Inh L3-6 VIP UG0898H09 Exc L5 THEMIS RGPD6 Exc L3-5 RORB LAMA4			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.4043	Mean : 0.6392	Mean : 0.5327	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :14.2709	Max. :15.6720	Max. :15.3644	
Exc L3-5 RORB LNX2 Micro L1-6 TYROBP CD74 Exc L5-6 FEZF2 FILIP1L			
Min. : 0.0000	Min. : 0.00000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.00000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.00000	Median : 0.0000	
Mean : 0.5852	Mean : 0.06015	Mean : 0.6228	
3rd Qu.: 0.0000	3rd Qu.: 0.00000	3rd Qu.: 0.0000	

Max. :15.4650	Max. :14.69370	Max. :15.5344	
Inh L1 LAMP5 NMBR	Exc L6 FEZF2 KLK7	Exc L5 FEZF2 PKD2L1	Inh L2-5 VIP SOX11
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.000
Mean : 0.3393	Mean : 0.5952	Mean : 0.5635	Mean : 0.478
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.000
Max. :14.4927	Max. :15.2798	Max. :15.7362	Max. :14.300
Exc L5 FEZF2 CSN1S1	Inh L3-5 SST CDH3	Inh L2 PVALB FRZB	
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.7327	Mean : 0.3475	Mean : 0.4476	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.8963	Max. :14.7563	Max. :14.9136	
Inh L5-6 PVALB SST CRHR2	Inh L1-2 VIP SCML4	Astro L1-6 FGFR3	AQP1
Min. : 0.0000	Min. : 0.0000	Min. : 0.00000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.00000	
Median : 0.0000	Median : 0.0000	Median : 0.00000	
Mean : 0.4702	Mean : 0.4986	Mean : 0.06774	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.00000	
Max. :15.5141	Max. :14.3126	Max. :14.98915	
Exc L2 LAMP5 KCNG3	Inh L1 LAMP5 BMP2	Inh L1-5 VIP CD27-AS1	Inh L1-3 VIP HSPB6
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4259	Mean : 0.3633	Mean : 0.3676	Mean : 0.2562
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.6062	Max. :14.6502	Max. :14.5148	Max. :14.9655
Inh L5-6 PVALB FAM150B	Inh L5-6 SST PIK3CD	Exc L6 FEZF2	POGK
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.4396	Mean : 0.3507	Mean : 0.5825	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.1584	Max. :14.5446	Max. :14.9563	
Inh L1-2 SST CLIC6	Inh L3-5 SST GGTLC3	Inh L3-5 VIP TAC3	Inh L5-6 SST ISX
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.3866	Mean : 0.2422	Mean : 0.2795	Mean : 0.3903
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.6960	Max. :14.9630	Max. :14.8366	Max. :15.0559
Exc L5-6 THEMIS	TNFAIP6	Oligo L2-6 OPALIN	FTH1P3
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.5412	Mean : 0.1224	Mean : 0.5684	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.3196	Max. :14.7106	Max. :15.2594	
Exc L5-6 FEZF2	C9orf135-AS1	Inh L1-3 VIP CHRNA2	Exc L3-5 RORB
			LINC01202

Min. : 0.0000	Min. : 0.0000	Min. : 0.00
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.00
Median : 0.0000	Median : 0.0000	Median : 0.00
Mean : 0.5219	Mean : 0.3905	Mean : 0.45
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.00
Max. :15.3415	Max. :14.7787	Max. :15.48

Inh L5 PVALB LRIG3 Exc L5-6 FEZF2 LPO Inh L1 LAMP5 PVRL2

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.5693	Mean : 0.6493	Mean : 0.3712
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.2553	Max. :15.2245	Max. :14.9122

Astro L1-6 FGFR3 PLCG1 Inh L2-5 PVALB HHIPL1 Inh L5-6 VIP COL4A3

Min. : 0.00000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.00000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.00000	Median : 0.0000	Median : 0.0000
Mean : 0.09598	Mean : 0.5386	Mean : 0.4585
3rd Qu.: 0.00000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.11265	Max. :15.3793	Max. :15.0196

Oligo L5-6 OPALIN LDLRAP1 Inh L5-6 PVALB KCNIP2 Exc L5 THEMIS LINC01116

Min. : 0.0000	Min. : 0.000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.000	Median : 0.0000
Mean : 0.1826	Mean : 0.574	Mean : 0.7489
3rd Qu.: 0.0000	3rd Qu.: 0.000	3rd Qu.: 0.0000
Max. :14.7400	Max. :15.284	Max. :15.8719

Inh L5-6 SST BEAN1 Exc L5-6 FEZF2 IFNG-AS1 Exc L3-5 FEZF2 ASGR2

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4187	Mean : 0.4989	Mean : 0.6781
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.6163	Max. :15.2586	Max. :15.7718

Exc L5 THEMIS VILL Inh L1 PAX6 MIR101-1 Inh L5-6 SST DNAJC14

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.6092	Mean : 0.4193	Mean : 0.5551
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.6660	Max. :15.1784	Max. :15.2494

Inh L3 PVALB SAMD13 Exc L3-5 FEZF2 LINC01107 Inh L5-6 SST C4orf26

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4452	Mean : 0.7116	Mean : 0.5385
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.3927	Max. :15.8796	Max. :15.1960

Inh L1 PVALB SST ASIC4 Inh L1-6 LAMP5 NES Inh L1-2 VIP PTGER3

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000

Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.2897	Mean : 0.2826	Mean : 0.4731	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :14.3122	Max. :14.9746	Max. :14.3524	
Inh L3-6 PAX6 LINC01497 Oligo L2-6 OPALIN MAP6D1 Inh L1 LAMP5 RAB11FIP1			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.4849	Mean : 0.4678	Mean : 0.4267	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.2055	Max. :14.9120	Max. :15.1065	
Exc L6 THEMIS SNTG2 Exc L3 RORB OTOGL Exc L6 FEZF2 PDYN			
Min. : 0.000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.000	Median : 0.0000	Median : 0.0000	
Mean : 0.604	Mean : 0.4608	Mean : 0.6197	
3rd Qu.: 0.000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :15.474	Max. :15.0690	Max. :15.3510	
OPC L1-6 PDGFRA COL20A1 Inh L5-6 PVALB ZFPM2-AS1 Inh L1 VIP KLHDC8B			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.1616	Mean : 0.5384	Mean : 0.3633	
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	
Max. :14.2120	Max. :15.2393	Max. :14.5433	
Inh L1-6 VIP SLC7A6OS Inh L1 SST DEFB108B Inh L5-6 SST PAWR Inh L2 PAX6 FREM2			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4398	Mean : 0.3037	Mean : 0.3618	Mean : 0.5384
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.1016	Max. :14.3231	Max. :14.6851	Max. :15.2546
Inh L1-2 SST PRRT4 Inh L3-5 PVALB ISG20 Inh L1-2 PVALB CDK20 Inh L1-2 VIP WNT4			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.000
Mean : 0.3557	Mean : 0.7167	Mean : 0.4745	Mean : 0.562
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.000
Max. :14.7855	Max. :15.1685	Max. :14.6697	Max. :14.735
Inh L5-6 LAMP5 CRABP1 Inh L1 SST P4HA3 Inh L1-3 VIP FNDC1 Inh L1-6 SST NPY			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4654	Mean : 0.3721	Mean : 0.4383	Mean : 0.4704
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.9060	Max. :14.3656	Max. :14.5632	Max. :15.4329
Oligo L3-6 OPALIN ENPP6 Inh L1-2 SST CCNJL Inh L1-5 VIP SMOC1			
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000	
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000	
Median : 0.0000	Median : 0.0000	Median : 0.0000	
Mean : 0.1996	Mean : 0.2905	Mean : 0.2806	

3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.8567	Max. :14.6491	Max. :14.8693
Exc L5 THEMIS SLC22A18 Inh L1-5 VIP LINC01013 Inh L2-5 VIP BSPRY		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.7461	Mean : 0.4034	Mean : 0.3476
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.4433	Max. :14.2684	Max. :15.1535
Inh L5-6 PVALB MEPE Exc L2-3 RORB CCDC68 Inh L5 SST RPL35AP11		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.4307	Mean : 0.5584	Mean : 0.3012
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.0504	Max. :15.3762	Max. :14.9717
Endo L2-5 NOSTRIN SRGN Inh L2-3 SST NMU Exc L6 THEMIS LINC00343		
Min. : 0.00000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.00000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.00000	Median : 0.0000	Median : 0.0000
Mean : 0.06623	Mean : 0.3941	Mean : 0.5737
3rd Qu.: 0.00000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.19969	Max. :14.9974	Max. :15.6847
Exc L5 THEMIS FGF10 Astro L1 FGFR3 SERPINI2 Inh L1-2 VIP EXPH5		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.6247	Mean : 0.1298	Mean : 0.3563
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.4664	Max. :15.3419	Max. :14.9797
Inh L1-5 VIP PHLDB3 Inh L5-6 SST KLHL1 Exc L5-6 FEZF2 CFTR		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.3241	Mean : 0.4385	Mean : 0.5596
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.1753	Max. :15.2324	Max. :15.5502
Exc L5-6 FEZF2 SH2D1B Inh L2-5 PVALB RPH3AL Exc L3 THEMIS ENPEP		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.6063	Mean : 0.4925	Mean : 0.5816
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :15.5240	Max. :15.2565	Max. :15.4214
Exc L6 FEZF2 PROKR2 Inh L1-2 VIP HTR3A Exc L5-6 FEZF2 OR1L8		
Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.6045	Mean : 0.3721	Mean : 0.6787
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.7011	Max. :14.2710	Max. :15.1655

Exc L2 LINC00507 GLRA3 Inh L3-5 SST OR5AH1P Exc L2-3 RORB RTKN2

Min. : 0.0000	Min. : 0.0000	Min. : 0.0000
1st Qu.: 0.0000	1st Qu.: 0.0000	1st Qu.: 0.0000
Median : 0.0000	Median : 0.0000	Median : 0.0000
Mean : 0.5073	Mean : 0.3633	Mean : 0.5916
3rd Qu.: 0.0000	3rd Qu.: 0.0000	3rd Qu.: 0.0000
Max. :14.6128	Max. :15.2374	Max. :14.6872

Exc L5 RORB MED8

Min. : 0.0000
 1st Qu.: 0.0000
 Median : 0.0000
 Mean : 0.5387
 3rd Qu.: 0.0000
 Max. :15.3911

[summary\(metadata\)](#)

sample_name	exp_component_name	cluster_label	cluster_color
Length:76533	Length:76533	Length:76533	Length:76533
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

cluster_order	class_label	class_color	class_order
Min. : 1.00	Length:76533	Length:76533	Min. :1.000
1st Qu.: 64.00	Class :character	Class :character	1st Qu.:1.000
Median : 77.00	Mode :character	Mode :character	Median :2.000
Mean : 73.21			Mean :1.739
3rd Qu.: 90.00			3rd Qu.:2.000
Max. :128.00			Max. :3.000

subclass_label	subclass_color	subclass_order	donor_sex_label
Length:76533	Length:76533	Min. : 1.000	Length:76533
Class :character	Class :character	1st Qu.: 6.000	Class :character
Mode :character	Mode :character	Median : 7.000	Mode :character
		Mean : 7.474	
		3rd Qu.: 8.000	
		Max. :20.000	

donor_sex_color	donor_sex_order	region_label	region_color
Length:76533	Min. :1.000	Length:76533	Length:76533
Class :character	1st Qu.:1.000	Class :character	Class :character
Mode :character	Median :1.000	Mode :character	Mode :character
	Mean :1.442		
	3rd Qu.:2.000		
	Max. :2.000		

region_order	cortical_layer_label	cortical_layer_color	cortical_layer_order
Min. :1	Length:76533	Length:76533	Min. :1
1st Qu.:1	Class :character	Class :character	1st Qu.:1
Median :1	Mode :character	Mode :character	Median :1
Mean :1			Mean :1

```

3rd Qu.:1                                3rd Qu.:1
Max.    :1                                Max.    :1
cell_type_accession_label cell_type_accession_color cell_type_accession_order
Length:76533                Length:76533                Min.    : 1.00
Class :character            Class :character            1st Qu.: 64.00
Mode  :character            Mode  :character            Median : 77.00
                                                Mean    : 73.21
                                                3rd Qu.: 90.00
                                                Max.    :128.00

cell_type_alias_label cell_type_alias_color cell_type_alias_order
Length:76533                Length:76533                Min.    : 1.00
Class :character            Class :character            1st Qu.: 64.00
Mode  :character            Mode  :character            Median : 77.00
                                                Mean    : 73.21
                                                3rd Qu.: 90.00
                                                Max.    :128.00

cell_type_alt_alias_label cell_type_alt_alias_color cell_type_alt_alias_order
Length:76533                Length:76533                Min.    : 1.00
Class :character            Class :character            1st Qu.: 64.00
Mode  :character            Mode  :character            Median : 77.00
                                                Mean    : 73.21
                                                3rd Qu.: 90.00
                                                Max.    :128.00

cell_type_designation_label cell_type_designation_color
Length:76533                Length:76533
Class :character            Class :character
Mode  :character            Mode  :character

cell_type_designation_order external_donor_name_label
Min.    : 1.00                Length:76533
1st Qu.: 64.00                Class :character
Median : 77.00                Mode  :character
Mean    : 73.21
3rd Qu.: 90.00
Max.    :128.00

external_donor_name_color external_donor_name_order specimen_type
Length:76533                Min.    :1.000                Length:76533
Class :character            1st Qu.:1.000                Class :character
Mode  :character            Median :1.000                Mode  :character
                                                Mean    :1.442
                                                3rd Qu.:2.000
                                                Max.    :2.000

full_genotype_label outlier_call outlier_type
Mode:logical                Mode :logical                Mode:logical
NA's:76533                FALSE:76533                NA's:76533

```

```
dim(trimmed_means_data)
```

```
[1] 50281 128
```

```
dim(metadata)
```

```
[1] 76533 39
```

```
# Removing zeros
meta_data_clean <- metadata %>%
  select(sample_name, cluster_label, class_label, subclass_label, donor_sex_label, region_label)

gene_zero_proportion <- colMeans(trimmed_means_data[, -1] == 0)

filtered_genes <- trimmed_means_data[, c(TRUE, gene_zero_proportion < 0.95)]

dim(filtered_genes)
```

```
[1] 50281 120
```

```
meta_data_clean %>%
  group_by(class_label) %>%
  summarise(count = n()) %>%
  ggplot(aes(x = class_label, y = count, fill = class_label)) +
  geom_bar(stat = "identity") +
  ggtitle("Class Distribution in Metadata") +
  theme_minimal()
```



```
meta_data_clean %>%
  group_by(subclass_label) %>%
  summarise(count = n()) %>%
  ggplot(aes(x = subclass_label, y = count, fill = subclass_label)) +
  geom_bar(stat = "identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ggtitle("Subclass Distribution") +
  theme_minimal()
```



```
# looking at 20 expressed genes
top_genes <- colMeans(filtered_genes[, -1]) %>%
```

```
sort(decreasing = TRUE) %>%
head(20)

# Plot
barplot(top_genes, las = 2, col = viridis::viridis(20), main = "Top 20 Expressed Genes")
```



Gene Expression Differences between Neuron Subclasses (SST vs. VIP)

H1: SST and VIP neurons will have distinct gene expression profiles.

```
meta_data_clean <- metadata %>%
  select(sample_name, cluster_label, class_label, subclass_label, donor_sex_label, region_label)

# SST and VIP neurons
sst_meta <- meta_data_clean %>% filter(subclass_label == "Sst")
vip_meta <- meta_data_clean %>% filter(subclass_label == "Vip")
```

```
dim(sst_meta)
```

```
[1] 5936    6
```

```
dim(vip_meta)
```

```
[1] 4858    6
```

```
head(trimmed_means_data)
```

```
# A tibble: 6 × 128
  feature      `Inh L1 PAX6 CHR` FAM7A` `Inh L1-3 VIP CBLN1` `Exc L3 LAMP5 CARM1P1`
  <chr>                <dbl>                <dbl>                <dbl>
1 DDX11L1              0              0              0
2 WASH7P               0              0              0
3 MIR6859-1            0              0              0
4 MIR1302-2            0              0              0
5 FAM138A              0              0              0
6 LOC1053792...        0              0              0
# i 124 more variables: `Inh L1-3 SST FAM20A` <dbl>,
#   `Inh L1-6 LAMP5 AARD` <dbl>, `Inh L1-6 LAMP5 CA1` <dbl>,
#   `Exc L6 THEMIS SLN` <dbl>, `Inh L2 VIP SLC6A16` <dbl>,
#   `Exc L5 FEZF2 NREP-AS1` <dbl>, `Inh L3-5 VIP IGDCC3` <dbl>,
#   `Inh L5-6 SST FBN2` <dbl>, `Inh L3-5 VIP HS3ST3A1` <dbl>,
```



```
# `Inh L1-6 PVALB COL15A1` <dbl>, `Exc L2-3 LINC00507 DSG3` <dbl>,
# `Exc L3-5 RORB TNNT2` <dbl>, `Exc L6 FEZF2 FFAR4` <dbl>, ...
```

```
# Extracting for SST and VIP neurons
sst_gene_expression <- trimmed_means_data %>%
  select(feature, contains("Sst"))

vip_gene_expression <- trimmed_means_data %>%
  select(feature, contains("Vip"))

dim(sst_gene_expression)
```

```
[1] 50281    24
```

```
dim(vip_gene_expression)
```

```
[1] 50281    25
```

```
head(sst_gene_expression)
```

```
# A tibble: 6 × 24
  feature      `Inh L1-3 SST FAM20A` `Inh L5-6 SST FBN2` `Inh L6 SST TH`
  <chr>                <dbl>          <dbl>          <dbl>
1 DDX11L1                0                0                0
2 WASH7P                  0                0                0
3 MIR6859-1              0                0                0
4 MIR1302-2              0                0                0
5 FAM138A                 0                0                0
6 LOC105379212           0                0                0
# i 20 more variables: `Inh L3-5 SST CDH3` <dbl>,
#   `Inh L5-6 PVALB SST CRHR2` <dbl>, `Inh L5-6 SST PIK3CD` <dbl>,
#   `Inh L1-2 SST CLIC6` <dbl>, `Inh L3-5 SST GGTLC3` <dbl>,
#   `Inh L5-6 SST ISX` <dbl>, `Inh L5-6 SST BEAN1` <dbl>,
#   `Inh L5-6 SST DNAJC14` <dbl>, `Inh L5-6 SST C4orf26` <dbl>,
#   `Inh L1 PVALB SST ASIC4` <dbl>, `Inh L1 SST DEFB108B` <dbl>,
#   `Inh L5-6 SST PAWR` <dbl>, `Inh L1-2 SST PRRT4` <dbl>, ...
```

```
head(vip_gene_expression)
```

```
# A tibble: 6 × 25
  feature      `Inh L1-3 VIP CBLN1` `Inh L2 VIP SLC6A16` `Inh L3-5 VIP IGDCC3`
  <chr>                <dbl>          <dbl>          <dbl>
1 DDX11L1                0                0                0
2 WASH7P                  0                0                0
3 MIR6859-1              0                0                0
4 MIR1302-2              0                0                0
5 FAM138A                 0                0                0
```

```
6 LOC105379212          0          0          0
# i 21 more variables: `Inh L3-5 VIP HS3ST3A1` <dbl>,
# `Inh L3-6 VIP ZIM2-AS1` <dbl>, `Inh L3-6 VIP UG0898H09` <dbl>,
# `Inh L2-5 VIP SOX11` <dbl>, `Inh L1-2 VIP SCML4` <dbl>,
# `Inh L1-5 VIP CD27-AS1` <dbl>, `Inh L1-3 VIP HSPB6` <dbl>,
# `Inh L3-5 VIP TAC3` <dbl>, `Inh L1-3 VIP CHRNA2` <dbl>,
# `Inh L5-6 VIP COL4A3` <dbl>, `Inh L1-2 VIP PTGER3` <dbl>,
# `Inh L1 VIP KLHDC8B` <dbl>, `Inh L1-6 VIP SLC7A60S` <dbl>, ...
```

```
# Filter for zero
sst_gene_expression_filtered <- sst_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

vip_gene_expression_filtered <- vip_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

dim(sst_gene_expression_filtered)
```

```
[1] 10359    24
```

```
dim(vip_gene_expression_filtered)
```

```
[1] 10246    25
```

```
head(sst_gene_expression_filtered)
```

```
# A tibble: 6 × 24
  feature `Inh L1-3 SST FAM20A` `Inh L5-6 SST FBN2` `Inh L6 SST TH`
  <chr>          <dbl>          <dbl>          <dbl>
1 LINC01128      0            0.644          0.581
2 NOC2L          0            0            0.594
3 HES4           0            2.25          3.34
4 AGRN           0            0            0.239
5 C1orf159       0            0            0.607
6 SDF4           0.0562        0.155          1.05
# i 20 more variables: `Inh L3-5 SST CDH3` <dbl>,
# `Inh L5-6 PVALB SST CRHR2` <dbl>, `Inh L5-6 SST PIK3CD` <dbl>,
# `Inh L1-2 SST CLIC6` <dbl>, `Inh L3-5 SST GGTLC3` <dbl>,
# `Inh L5-6 SST ISX` <dbl>, `Inh L5-6 SST BEAN1` <dbl>,
# `Inh L5-6 SST DNAJC14` <dbl>, `Inh L5-6 SST C4orf26` <dbl>,
# `Inh L1 PVALB SST ASIC4` <dbl>, `Inh L1 SST DEFB108B` <dbl>,
# `Inh L5-6 SST PAWR` <dbl>, `Inh L1-2 SST PRRT4` <dbl>, ...
```

```
head(vip_gene_expression_filtered)
```

```
# A tibble: 6 × 25
  feature `Inh L1-3 VIP CBLN1` `Inh L2 VIP SLC6A16` `Inh L3-5 VIP IGDCC3`
```

	<chr>	<dbl>	<dbl>	<dbl>
1	LINC01128	0	0	0
2	NOC2L	0	0	0
3	HES4	0	0.614	0
4	C1orf159	0	0.301	0.0191
5	SDF4	0.328	0.145	0.451
6	UBE2J2	0	0	0

```
# i 21 more variables: `Inh L3-5 VIP HS3ST3A1` <dbl>,
# `Inh L3-6 VIP ZIM2-AS1` <dbl>, `Inh L3-6 VIP UG0898H09` <dbl>,
# `Inh L2-5 VIP SOX11` <dbl>, `Inh L1-2 VIP SCML4` <dbl>,
# `Inh L1-5 VIP CD27-AS1` <dbl>, `Inh L1-3 VIP HSPB6` <dbl>,
# `Inh L3-5 VIP TAC3` <dbl>, `Inh L1-3 VIP CHRNA2` <dbl>,
# `Inh L5-6 VIP COL4A3` <dbl>, `Inh L1-2 VIP PTGER3` <dbl>,
# `Inh L1 VIP KLHDC8B` <dbl>, `Inh L1-6 VIP SLC7A60S` <dbl>, ...
```

```
sst_gene_expression_filtered <- sst_gene_expression_filtered[rowSums(select(sst_gene_expression_f:
vip_gene_expression_filtered <- vip_gene_expression_filtered[rowSums(select(vip_gene_expression_f:
```

```
# t-test (Had to modify the t test code from class using sourcegraph because that t-test was comi
t_test_results <- lapply(1:nrow(sst_gene_expression_filtered), function(i) {
```

```
  sst_values <- as.numeric(sst_gene_expression_filtered[i, -1])
  vip_values <- as.numeric(vip_gene_expression_filtered[i, -1])
```

```
  if (length(sst_values) > 1 && length(vip_values) > 1) {
    t_res <- tryCatch(
      t.test(sst_values, vip_values),
      error = function(e) NULL
    )

    if (!is.null(t_res)) {
      return(data.frame(
        Gene = sst_gene_expression_filtered$feature[i],
        t_value = round(t_res$statistic, 2),
        df = round(t_res$parameter, 2),
        p_value = ifelse(t_res$p.value < 0.001, "< .001", round(t_res$p.value, 3)),
        conf_int_low = round(t_res$conf.int[1], 2),
        conf_int_high = round(t_res$conf.int[2], 2),
        mean_diff = round(t_res$estimate[1] - t_res$estimate[2], 2)
      ))
    }
  }
  return(NULL)
})
```

```
# Combining
t_test_df <- do.call(rbind, t_test_results)
```

```
# Formatting
```

```
t_test_df$APA_Format <- with(t_test_df, paste0(
  "t(", df, ") = ", t_value,
  ", p = ", p_value,
  ", 95% CI [", conf_int_low, ", ", conf_int_high,
  "], mean difference = ", mean_diff
))

significant_genes <- t_test_df %>%
  filter(as.numeric(gsub("< .001", "0.001", p_value)) < 0.05) %>%
  arrange(as.numeric(gsub("< .001", "0.001", p_value)))
print(head(significant_genes))
```

	Gene	t_value	df	p_value	conf_int_low	conf_int_high	mean_diff
t6	UBE2J2	-12.19	33.52	< .001	-1.52	-1.08	-1.30
t7	ACAP3	5.39	42.17	< .001	0.56	1.23	0.90
t8	CPSF3L	5.50	22.59	< .001	0.31	0.69	0.50
t14	SSU72	4.87	22.42	< .001	0.46	1.13	0.80
t17	SLC35E2	-20.22	23.19	< .001	-6.67	-5.43	-6.05
t18	GNB1	11.37	28.18	< .001	3.44	4.96	4.20

APA_Format

```
t6  t(33.52) = -12.19, p = < .001, 95% CI [-1.52, -1.08], mean difference = -1.3
t7      t(42.17) = 5.39, p = < .001, 95% CI [0.56, 1.23], mean difference = 0.9
t8      t(22.59) = 5.5, p = < .001, 95% CI [0.31, 0.69], mean difference = 0.5
t14     t(22.42) = 4.87, p = < .001, 95% CI [0.46, 1.13], mean difference = 0.8
t17 t(23.19) = -20.22, p = < .001, 95% CI [-6.67, -5.43], mean difference = -6.05
t18     t(28.18) = 11.37, p = < .001, 95% CI [3.44, 4.96], mean difference = 4.2
```

```
top5_genes <- significant_genes %>%
  mutate(p_value = as.numeric(gsub("< .001", "0.001", p_value))) %>%
  arrange(p_value) %>%
  head(5)

print(top5_genes)
```

	Gene	t_value	df	p_value	conf_int_low	conf_int_high	mean_diff
t6	UBE2J2	-12.19	33.52	0.001	-1.52	-1.08	-1.30
t7	ACAP3	5.39	42.17	0.001	0.56	1.23	0.90
t8	CPSF3L	5.50	22.59	0.001	0.31	0.69	0.50
t14	SSU72	4.87	22.42	0.001	0.46	1.13	0.80
t17	SLC35E2	-20.22	23.19	0.001	-6.67	-5.43	-6.05

APA_Format

```
t6  t(33.52) = -12.19, p = < .001, 95% CI [-1.52, -1.08], mean difference = -1.3
t7      t(42.17) = 5.39, p = < .001, 95% CI [0.56, 1.23], mean difference = 0.9
t8      t(22.59) = 5.5, p = < .001, 95% CI [0.31, 0.69], mean difference = 0.5
t14     t(22.42) = 4.87, p = < .001, 95% CI [0.46, 1.13], mean difference = 0.8
t17 t(23.19) = -20.22, p = < .001, 95% CI [-6.67, -5.43], mean difference = -6.05
```

```
print(head(t_test_df$APA_Format))
```

```
[1] "t(29.85) = 1.21, p = 0.235, 95% CI [-0.06, 0.24], mean difference = 0.09"
[2] "t(40.06) = 0.26, p = 0.792, 95% CI [-0.11, 0.15], mean difference = 0.02"
[3] "t(29.84) = 2.46, p = 0.02, 95% CI [0.17, 1.85], mean difference = 1.01"
[4] "t(37.35) = -2.61, p = 0.013, 95% CI [-0.59, -0.07], mean difference = -0.33"
[5] "t(42.41) = -2.43, p = 0.019, 95% CI [-0.68, -0.06], mean difference = -0.37"
[6] "t(43.04) = 2.61, p = 0.013, 95% CI [0.06, 0.49], mean difference = 0.28"
```

The t-test analysis found that some genes showed significant differences in their expression between SST and VIP neurons, while others did not. One gene that stood out the most was SLC35E2 because it showed a clear difference in expression between the neuron types ($t(23.19) = -20.22$, $p < .001$). The average difference in expression was -6.67, with a 95% confidence interval between -6.67 and -5.43. This indicates that SLC35E2 is significantly less expressed in one neuron type compared to the other.

Some other genes also showed substantial differences. For instance, GNB1 had a marked difference in expression ($t(28.18) = 11.37$, $p < .001$) with an average difference of 3.44 (95% CI [3.44, 4.96]), indicating it is significantly more expressed in one neuron type. Similarly, UBE2J2 displayed a significant difference ($t(33.52) = -12.19$, $p < .001$), with an average difference of -1.30 (95% CI [-1.52, -1.08]), suggesting it is less expressed in one of the neuron types.

Other significant genes included ACAP3, CPSF3L, and SSU72, which also showed clear differences. ACAP3 had a positive difference ($t(42.17) = 5.39$, $p < .001$), with an average difference of 0.90 (95% CI [0.56, 1.23]). CPSF3L also had a positive difference ($t(22.59) = 5.50$, $p < .001$), with an average difference of 0.50 (95% CI [0.31, 0.69]). Meanwhile, SSU72 showed a difference ($t(22.42) = 4.87$, $p < .001$), with an average difference of 0.80 (95% CI [0.46, 1.13]).

In summary, while genes like SLC35E2, GNB1, UBE2J2, ACAP3, CPSF3L, and SSU72 show significant differences in expression between SST and VIP neurons, the variability among them highlights how distinct gene expression can be across different neuron subclasses.

Plot for H1 (Bonx Plot)

```
# SST
sst_top5 <- sst_gene_expression_filtered %>%
  filter(feature %in% top5_genes$Gene) %>%
  pivot_longer(cols = -feature, names_to = "Sample", values_to = "Expression") %>%
  mutate(Type = "SST")

# VIP
vip_top5 <- vip_gene_expression_filtered %>%
  filter(feature %in% top5_genes$Gene) %>%
  pivot_longer(cols = -feature, names_to = "Sample", values_to = "Expression") %>%
  mutate(Type = "VIP")

# Combine
boxplot_data <- bind_rows(sst_top5, vip_top5)
```

```
# Boxplot
ggplot(boxplot_data, aes(x = feature, y = Expression, fill = Type)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.6) +
  labs(title = "Expression Levels of Top 5 Significant Genes (SST vs VIP)",
       x = "Gene", y = "Expression Level") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



I chose a boxplot for H1 because it clearly shows how gene expression levels differ between SST and VIP neurons when it comes to variability between the two groups.

The boxplot shows the expression levels of the top 5 significant genes (ACAP3, CPSF3L, SLC35E2, SSU72, and UBE2J2) between SST and VIP neurons. The x-axis lists the gene names, while the y-axis shows their expression levels. Red represents SST neurons, and blue represents VIP neurons.

Looking at ACAP3, we see that VIP neurons generally show slightly higher expression levels compared to SST neurons. This matches the t-test result, where ACAP3 showed a positive difference ($t(42.17) = 5.39$, $p < .001$), indicating that it is more expressed in VIP neurons.

The plot shows that ACAP3 and CPSF3L are more expressed in VIP neurons, while SLC35E2, SSU72, and UBE2J2 are more expressed in SST neurons, matching the t-test results.

Overall, the plot aligns well with the t-test results, showing consistent differences between SST and VIP neurons for each gene. The variability between neuron types is clearly visualized, reinforcing the differences detected statistically.

Influence of Donor Sex on Gene Expression in Specific Neuron Types

H2: Donor sex may lead to small but noticeable differences in gene expression within some cell types.

```
# Extracting gene expression for male and female
female_gene_expression <- trimmed_means_data %>%
  select(feature, contains("F"))

male_gene_expression <- trimmed_means_data %>%
  select(feature, contains("M"))

#Removing zeroes

female_gene_expression_filtered <- female_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

male_gene_expression_filtered <- male_gene_expression %>%
```

```
filter(rowSums(select(., -feature)) > 0)
```

```
dim(female_gene_expression_filtered)
```

```
[1] 13354    38
```

```
dim(male_gene_expression_filtered)
```

```
[1] 13052    40
```

```
head(female_gene_expression_filtered)
```

```
# A tibble: 6 × 38
```

	feature	`Inh L1 PAX6 CHR FAM7A`	`Inh L1-3 SST FAM20A`	Exc L5 FEZF2 NREP-AS... ¹
	<chr>	<dbl>	<dbl>	<dbl>
1	LOC102723...	0	0	0
2	LOC105378...	0	0	0
3	LINC01128	0	0	0.114
4	NOC2L	0	0	0.203
5	HES4	0.126	0	0
6	AGRN	0	0	0.819

```
# i abbreviated name: 1`Exc L5 FEZF2 NREP-AS1`
```

```
# i 34 more variables: `Inh L5-6 SST FBN2` <dbl>, `Exc L6 FEZF2 FFAR4` <dbl>,
```

```
# `VLMC L1-5 PDGFRA COLEC12` <dbl>, `Exc L5 FEZF2 RNF144A-AS1` <dbl>,
```

```
# `Exc L5-6 FEZF2 FILIP1L` <dbl>, `Exc L6 FEZF2 KLK7` <dbl>,
```

```
# `Exc L5 FEZF2 PKD2L1` <dbl>, `Exc L5 FEZF2 CSN1S1` <dbl>,
```

```
# `Inh L2 PVALB FRZB` <dbl>, `Astro L1-6 FGFR3 AQP1` <dbl>,
```

```
# `Inh L5-6 PVALB FAM150B` <dbl>, `Exc L6 FEZF2 POGK` <dbl>, ...
```

```
head(male_gene_expression_filtered)
```

```
# A tibble: 6 × 40
```

	feature	`Inh L1 PAX6 CHR FAM7A`	`Exc L3 LAMP5 CARM1P1`	`Inh L1-3 SST FAM20A`
	<chr>	<dbl>	<dbl>	<dbl>
1	LOC102723...	0	0	0
2	LINC01128	0	1.13	0
3	NOC2L	0	0.516	0
4	HES4	0.126	0	0
5	AGRN	0	2.11	0
6	C1orf159	0	1.84	0

```
# i 36 more variables: `Inh L1-6 LAMP5 AARD` <dbl>, `Inh L1-6 LAMP5 CA1` <dbl>,
```

```
# `Exc L6 THEMIS SLN` <dbl>, `Inh L3-6 VIP ZIM2-AS1` <dbl>,
```

```
# `Exc L5-6 THEMIS SMYD1` <dbl>, `VLMC L1-5 PDGFRA COLEC12` <dbl>,
```

```
# `Exc L5 THEMIS RGPD6` <dbl>, `Exc L3-5 RORB LAMA4` <dbl>,
```

```
# `Micro L1-6 TYROBP CD74` <dbl>, `Inh L1 LAMP5 NMBR` <dbl>,
```

```
# `Inh L1-2 VIP SCML4` <dbl>, `Exc L2 LAMP5 KCNG3` <dbl>,
```

```
# `Inh L1 LAMP5 BMP2` <dbl>, `Inh L5-6 PVALB FAM150B` <dbl>, ...
```

```

# Two-Sample T-Test
t_test_results <- list()
count <- 0

for (i in 1:nrow(female_gene_expression_filtered)) {

  female_values <- as.numeric(female_gene_expression_filtered[i, -1])
  male_values <- as.numeric(male_gene_expression_filtered[i, -1])

  # NA values
  female_values <- female_values[!is.na(female_values)]
  male_values <- male_values[!is.na(male_values)]

  if (length(female_values) >= 2 && length(male_values) >= 2) {

    t_test <- t.test(female_values, male_values)

    count <- count + 1
    t_test_results[[count]] <- data.frame(
      Gene = female_gene_expression_filtered$feature[i],
      p_value = t_test$p.value,
      mean_female = mean(female_values),
      mean_male = mean(male_values),
      difference = mean(female_values) - mean(male_values)
    )
  }
}

all_results <- do.call(rbind, t_test_results)

# significant genes (p < 0.05)
significant_genes <- all_results[all_results$p_value < 0.05, ]

# Sort
significant_genes <- significant_genes[order(significant_genes$p_value), ]

print(head(significant_genes))

```

	Gene	p_value	mean_female	mean_male	difference
12804	TAB1	7.624439e-96	0.07399813	10.4257462	-10.351748
5441	MAGI2	2.494417e-71	11.07369961	0.4121887	10.661511
67	ERRFI1	1.504860e-64	0.54335358	8.6958068	-8.152453
1205	IRF2BP2	7.211101e-63	0.70481118	9.2888147	-8.584004
5543	LAMTOR4	2.192298e-59	0.42420875	8.3894079	-7.965199
12791	DDX17	1.872627e-58	8.02759240	0.7046707	7.322922


```
print(paste("Found", nrow(significant_genes), "significant genes"))
```

```
[1] "Found 10252 significant genes"
```

The test found that some genes do have significantly different expression levels between male and female samples. For example, the gene TAB1 is expressed much more in males (average = 10.43) compared to females (average = 0.07), with a big difference of about 10.35 ($p < .001$). On the other hand, the gene MAGI2 shows the opposite pattern, being higher in females (average = 11.07) compared to males (average = 0.41), with a difference of about 10.66 ($p < .001$).

Other genes like ERFFI1, IRF2BP2, LAMTOR4, and DDX17 also show significant differences between the sexes, with p-values less than .001. This means that these genes might be influenced by sex.

Plot -H2 (Double Bar graph)

```
top_genes <- significant_genes %>%
  head(5)

ggplot(top_genes, aes(x = Gene)) +
  geom_bar(aes(y = mean_female, fill = "Female"), stat = "identity", position = "dodge", color = "black") +
  geom_bar(aes(y = mean_male, fill = "Male"), stat = "identity", position = "dodge", color = "black") +
  scale_fill_manual(values = c("Female" = "blue", "Male" = "red")) +
  labs(title = "Mean Gene Expression Levels: Male vs Female",
       x = "Gene",
       y = "Mean Expression Level",
       fill = "Sex") +
  theme_minimal()
```



This graph above shows a comparison of the average gene expression levels for the top five significant genes between male and female samples. The x-axis shows the genes while the y axis shows the expression levels .

Just like the t-test, we see that the gene MAGI2 shows much higher expression in females compared to males, while the other four genes (ERFFI1, IRF2BP2, LAMTOR4, and TAB1) are expressed more in males. This difference in expression supports my hypothesis that genes may indeed be regulated differently between males and females, with MAGI2

Originally, I tried to use a boxplot to visualize these differences, but the boxplot didn't work well because it appeared almost flat, showing little to no variation. This was most likely because the data lacked variability within each gene-sex group. In order to make the differences clearer, I switched to a double bar plot, which more directly shows the average expression levels and makes it easier to see the contrasts between male and female samples.

Gene Expression Variability Across Cortical Layers

H3: Genes involved in neurotransmission (like synaptic communication) will vary across different cortical layers.

```
# GABAergic, Glutamatergic markers
neuro_genes <- c("GABRA1", "GABRG2", "GRIN1", "GRIA1", "SLC17A7") # Example genes

# Extraction
neuro_gene_expression <- trimmed_means_data %>%
  filter(feature %in% neuro_genes)

dim(neuro_gene_expression)
```

```
[1] 5 128
```

```
head(neuro_gene_expression)
```

```
# A tibble: 5 × 128
  feature `Inh L1 PAX6 CHRFAM7A` `Inh L1-3 VIP CBLN1` `Exc L3 LAMP5 CARM1P1`
  <chr>          <dbl>          <dbl>          <dbl>
1 GRIA1          9.52          7.17          7.39
2 GABRA1          8.86          7.85          7.96
3 GABRG2          8.79          7.49          8.16
4 GRIN1          7.16          7.29          7.50
5 SLC17A7         0            0            6.33
# i 124 more variables: `Inh L1-3 SST FAM20A` <dbl>,
# `Inh L1-6 LAMP5 AARD` <dbl>, `Inh L1-6 LAMP5 CA1` <dbl>,
# `Exc L6 THEMIS SLN` <dbl>, `Inh L2 VIP SLC6A16` <dbl>,
# `Exc L5 FEZF2 NREP-AS1` <dbl>, `Inh L3-5 VIP IGDCC3` <dbl>,
# `Inh L5-6 SST FBN2` <dbl>, `Inh L3-5 VIP HS3ST3A1` <dbl>,
# `Inh L1-6 PVALB COL15A1` <dbl>, `Exc L2-3 LINC00507 DSG3` <dbl>,
# `Exc L3-5 RORB TNNT2` <dbl>, `Exc L6 FEZF2 FFAR4` <dbl>, ...
```

```
# Extract for cortical L
layer_labels <- gsub(".*L(\\d+).*", "L\\1", colnames(neuro_gene_expression)[-1])

# Combine gene expression data with layer labels
neuro_gene_expression_long <- neuro_gene_expression %>%
  pivot_longer(cols = -feature, names_to = "sample", values_to = "expression") %>%
  mutate(layer = layer_labels[match(sample, colnames(neuro_gene_expression)[-1])])

head(neuro_gene_expression_long)
```

```
# A tibble: 6 × 4
  feature sample          expression layer
  <chr>   <chr>          <dbl> <chr>
1 GRIA1   Inh L1 PAX6 CHRFAM7A    9.52 L1
```

2	GRIA1	Inh	L1-3	VIP	CBLN1	7.17	L1
3	GRIA1	Exc	L3	LAMP5	CARM1P1	7.39	L3
4	GRIA1	Inh	L1-3	SST	FAM20A	10.1	L1
5	GRIA1	Inh	L1-6	LAMP5	AARD	9.35	L1
6	GRIA1	Inh	L1-6	LAMP5	CA1	10.3	L1

```
# I went with ANOVA here instead of a t-test because ANOVA allows us to compare the means across

# ANOVA (code structure provided by stack overflow)
anova_results <- neuro_gene_expression_long %>%
  group_by(feature) %>%
  summarise(
    anova = list(aov(expression ~ layer)),
    p_value = summary(aov(expression ~ layer))[[1]][["Pr(>F)"]][1]
  ) %>%
  filter(p_value < 0.05) %>%
  arrange(p_value)

print(head(anova_results))
```

```
# A tibble: 2 × 3
  feature anova      p_value
  <chr>   <list>    <dbl>
1 SLC17A7 <aov> 0.000000115
2 GRIN1   <aov> 0.0357
```

```
print(paste("Number of significant genes:", nrow(anova_results)))
```

```
[1] "Number of significant genes: 2"
```

The ANOVA analysis showed that there are significant differences in gene expression across cortical layers for some key neurotransmission genes. For instance, the gene SLC17A7 showed a significant difference in expression between layers, $F(6, 121) = [\text{calculated } F \text{ value}]$, $p < .001$. Similarly, GRIN1 also demonstrated significant variation across layers, $F(6, 121) = [\text{calculated } F \text{ value}]$, $p < .05$. These results suggest that gene expression for neurotransmission-related genes varies depending on the cortical layer, indicating layer-specific roles in neurotransmission.

Graph -H3 (Box Plot)

```
top_genes <- c("SLC17A7", "GRIN1")

plot_data <- neuro_gene_expression_long %>%
  filter(feature %in% top_genes)
```

```
# box plot
ggplot(plot_data, aes(x = layer, y = expression, fill = feature)) +
  geom_boxplot() +
  labs(title = "Gene Expression Across Cortical Layers",
       x = "Cortical Layer",
       y = "Gene Expression Level",
       fill = "Gene") +
  theme_minimal()
```



This box plot shows the gene expression levels of two key neurotransmission-related genes, GRIN1 and SLC17A7, across different cortical layers. Each box represents the distribution of gene expression within a specific layer, and the colors differentiate between the two genes: red for GRIN1 and teal for SLC17A7.

Looking at the plot, we can see that the GRIN1 expression tends to be more consistent across layers, with noticeable expression in L1, L3, L5, and L6. On the other hand, SLC17A7 has a more varied pattern, with high expression in L2 and L3, but significantly lower levels in other layers. There's also some variability within each layer, as indicated by the spread of the boxes and the presence of a few outliers.

This graph makes it clear that while both genes are involved in neurotransmission, they don't follow the same expression pattern across the cortical layers, highlighting potential differences in their roles or regulation.

Hypothesis 4: Specific Marker Genes and Major Cell Classes

H4: Specific marker genes will be strongly associated with GABAergic, Glutamatergic, or Non-neuronal cell classes.

```
# Extract GABAergic neurons (Inhibitory)
gabaergic_gene_expression <- trimmed_means_data %>%
  select(feature, contains("Inh"))

# Extract Glutamatergic neurons (Excitatory)
glutamatergic_gene_expression <- trimmed_means_data %>%
  select(feature, contains("Exc"))

# Extract Non-neuronal cells
non_neuronal_gene_expression <- trimmed_means_data %>%
  select(feature, contains("Astro"), contains("Oligo"), contains("Micro"), contains("OPC"), contains("Epend"))

dim(gabaergic_gene_expression)
```

[1] 50281 73

```
dim(glutamatergic_gene_expression)
```

```
[1] 50281    45
```

```
dim(non_neuronal_gene_expression)
```

```
[1] 50281    12
```

```
head(gabaergic_gene_expression)
```

```
# A tibble: 6 × 73
  feature      `Inh L1 PAX6 CHR FAM7A` `Inh L1-3 VIP CBLN1` `Inh L1-3 SST FAM20A`
  <chr>                <dbl>                <dbl>                <dbl>
1 DDX11L1              0              0              0
2 WASH7P               0              0              0
3 MIR6859-1            0              0              0
4 MIR1302-2            0              0              0
5 FAM138A              0              0              0
6 LOC105379212         0              0              0
# i 69 more variables: `Inh L1-6 LAMP5 AARD` <dbl>, `Inh L1-6 LAMP5 CA1` <dbl>,
#   `Inh L2 VIP SLC6A16` <dbl>, `Inh L3-5 VIP IGDCC3` <dbl>,
#   `Inh L5-6 SST FBN2` <dbl>, `Inh L3-5 VIP HS3ST3A1` <dbl>,
#   `Inh L1-6 PVALB COL15A1` <dbl>, `Inh L3-6 VIP ZIM2-AS1` <dbl>,
#   `Inh L5-6 PVALB GAPDHP60` <dbl>, `Inh L6 SST TH` <dbl>,
#   `Inh L3-6 VIP UG0898H09` <dbl>, `Inh L1 LAMP5 NMBR` <dbl>,
#   `Inh L2-5 VIP SOX11` <dbl>, `Inh L3-5 SST CDH3` <dbl>, ...
```

```
head(glutamatergic_gene_expression)
```

```
# A tibble: 6 × 45
  feature      `Exc L3 LAMP5 CARM1P1` `Exc L6 THEMIS SLN` `Exc L5 FEZF2 NREP-AS...`1
  <chr>                <dbl>                <dbl>                <dbl>
1 DDX11L1              0              0              0
2 WASH7P               0              0              0
3 MIR6859-1            0              0              0
4 MIR1302-2            0              0              0
5 FAM138A              0              0              0
6 LOC105379212         0              0              0
# i abbreviated name: 1`Exc L5 FEZF2 NREP-AS1`
# i 41 more variables: `Exc L2-3 LINC00507 DSG3` <dbl>,
#   `Exc L3-5 RORB TNNT2` <dbl>, `Exc L6 FEZF2 FFAR4` <dbl>,
#   `Exc L2 LINC00507 ATP7B` <dbl>, `Exc L5-6 THEMIS SMYD1` <dbl>,
#   `Exc L2-3 RORB PTPN3` <dbl>, `Exc L5 FEZF2 RNF144A-AS1` <dbl>,
#   `Exc L5 THEMIS RGPD6` <dbl>, `Exc L3-5 RORB LAMA4` <dbl>,
#   `Exc L3-5 RORB LNX2` <dbl>, `Exc L5-6 FEZF2 FILIP1L` <dbl>, ...
```

```
head(non_neuronal_gene_expression)
```

```
# A tibble: 6 × 12
  feature      Astro L1-6 FGFR3 AQP...1 Astro L1-6 FGFR3 PLC...2 Astro L1 FGFR3 SERPI...3
```

	<chr>	<dbl>	<dbl>	<dbl>
1	DDX11L1	0	0	0
2	WASH7P	0	0	0
3	MIR6859-1	0	0	0
4	MIR1302-2	0	0	0
5	FAM138A	0	0	0
6	LOC10537...	0	0	0

```

# i abbreviated names: 1`Astro L1-6 FGFR3 AQP1`, 2`Astro L1-6 FGFR3 PLCG1`,
#   3`Astro L1 FGFR3 SERPINI2`
# i 8 more variables: `Oligo L2-6 OPALIN FTH1P3` <dbl>,
#   `Oligo L5-6 OPALIN LDLRAP1` <dbl>, `Oligo L2-6 OPALIN MAP6D1` <dbl>,
#   `Oligo L3-6 OPALIN ENPP6` <dbl>, `Micro L1-6 TYROBP CD74` <dbl>,
#   `OPC L1-6 PDGFRA COL20A1` <dbl>, `Endo L2-5 NOSTRIN SRGN` <dbl>,
#   `VLMC L1-5 PDGFRA COLEC12` <dbl>

```

```

# Zero
gabaergic_gene_expression_filtered <- gabaergic_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

glutamatergic_gene_expression_filtered <- glutamatergic_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

non_neuronal_gene_expression_filtered <- non_neuronal_gene_expression %>%
  filter(rowSums(select(., -feature)) > 0)

dim(gabaergic_gene_expression_filtered)

```

```
[1] 12434    73
```

```
dim(glutamatergic_gene_expression_filtered)
```

```
[1] 13146    45
```

```
dim(non_neuronal_gene_expression_filtered)
```

```
[1] 8489    12
```

```
head(gabaergic_gene_expression_filtered)
```

```

# A tibble: 6 × 73
  feature      `Inh L1 PAX6 CHRFAM7A` `Inh L1-3 VIP CBLN1` `Inh L1-3 SST FAM20A`
  <chr>          <dbl>          <dbl>          <dbl>
1 LOC102723917      0              0              0
2 LINC01128          0              0              0
3 NOC2L             0              0              0
4 KLHL17             0              0              0
5 HES4              0.126          0              0

```

```

6 AGRN                                0                                0                                0
# i 69 more variables: `Inh L1-6 LAMP5 AARD` <dbl>, `Inh L1-6 LAMP5 CA1` <dbl>,
#   `Inh L2 VIP SLC6A16` <dbl>, `Inh L3-5 VIP IGDCC3` <dbl>,
#   `Inh L5-6 SST FBN2` <dbl>, `Inh L3-5 VIP HS3ST3A1` <dbl>,
#   `Inh L1-6 PVALB COL15A1` <dbl>, `Inh L3-6 VIP ZIM2-AS1` <dbl>,
#   `Inh L5-6 PVALB GAPDHP60` <dbl>, `Inh L6 SST TH` <dbl>,
#   `Inh L3-6 VIP UG0898H09` <dbl>, `Inh L1 LAMP5 NMBR` <dbl>,
#   `Inh L2-5 VIP SOX11` <dbl>, `Inh L3-5 SST CDH3` <dbl>, ...

```

```
head(glutamatergic_gene_expression_filtered)
```

```

# A tibble: 6 × 45
  feature      `Exc L3 LAMP5 CARM1P1` `Exc L6 THEMIS SLN` Exc L5 FEZF2 NREP-AS...1
  <chr>                <dbl>                <dbl>                <dbl>
1 LOC102723917          0                  0                  0
2 LOC105378582          0                  0                  0
3 LINC01128             1.13             0.686             0.114
4 NOC2L                 0.516            0.189             0.203
5 HES4                  0                  0                  0
6 AGRN                  2.11             1.23             0.819
# i abbreviated name: 1`Exc L5 FEZF2 NREP-AS1`
# i 41 more variables: `Exc L2-3 LINC00507 DSG3` <dbl>,
#   `Exc L3-5 RORB TNNT2` <dbl>, `Exc L6 FEZF2 FFAR4` <dbl>,
#   `Exc L2 LINC00507 ATP7B` <dbl>, `Exc L5-6 THEMIS SMYD1` <dbl>,
#   `Exc L2-3 RORB PTPN3` <dbl>, `Exc L5 FEZF2 RNF144A-AS1` <dbl>,
#   `Exc L5 THEMIS RGPD6` <dbl>, `Exc L3-5 RORB LAMA4` <dbl>,
#   `Exc L3-5 RORB LNX2` <dbl>, `Exc L5-6 FEZF2 FILIP1L` <dbl>, ...

```

```
head(non_neuronal_gene_expression_filtered)
```

```

# A tibble: 6 × 12
  feature `Astro L1-6 FGFR3 AQP1` Astro L1-6 FGFR3 PLC...1 Astro L1 FGFR3 SERPI...2
  <chr>                <dbl>                <dbl>                <dbl>
1 AGRN                  0                  0                  0
2 C1orf159              0                  0                  0
3 SDF4                  0                  0                  0
4 ACAP3                 0                  0                  0
5 CPSF3L                0                  0                  0
6 CCNL2                 1.17             1.03             0.936
# i abbreviated names: 1`Astro L1-6 FGFR3 PLCG1`, 2`Astro L1 FGFR3 SERPINI2`
# i 8 more variables: `Oligo L2-6 OPALIN FTH1P3` <dbl>,
#   `Oligo L5-6 OPALIN LDLRAP1` <dbl>, `Oligo L2-6 OPALIN MAP6D1` <dbl>,
#   `Oligo L3-6 OPALIN ENPP6` <dbl>, `Micro L1-6 TYROBP CD74` <dbl>,
#   `OPC L1-6 PDGFRA COL20A1` <dbl>, `Endo L2-5 NOSTRIN SRGN` <dbl>,
#   `VLMC L1-5 PDGFRA COLEC12` <dbl>

```

```

# ANOVA ( code template by Stack overflow)
anova_results <- list()

```

```

for (i in 1:nrow(gabaergic_gene_expression_filtered)) {
  gene <- gabaergic_gene_expression_filtered$feature[i]

  # Combine the expression data from the three classes
  expression_values <- c(
    as.numeric(gabaergic_gene_expression_filtered[i, -1]),
    as.numeric(glutamatergic_gene_expression_filtered[i, -1]),
    as.numeric(non_neuronal_gene_expression_filtered[i, -1])
  )

  group_labels <- c(
    rep("GABAergic", ncol(gabaergic_gene_expression_filtered) - 1),
    rep("Glutamatergic", ncol(glutamatergic_gene_expression_filtered) - 1),
    rep("Non-neuronal", ncol(non_neuronal_gene_expression_filtered) - 1)
  )

  # NA values
  valid <- !is.na(expression_values)
  expression_values <- expression_values[valid]
  group_labels <- group_labels[valid]

  anova_result <- aov(expression_values ~ group_labels)

  p_value <- summary(anova_result)[[1]][["Pr(>F)"]][1]

  anova_results[[i]] <- data.frame(
    Gene = gene,
    p_value = p_value
  )
}

anova_df <- do.call(rbind, anova_results)

# p < 0.05
significant_genes <- anova_df %>% filter(p_value < 0.05)

significant_genes <- significant_genes %>% arrange(p_value)

print(head(significant_genes))

```

	Gene	p_value
1	NUP35	2.620465e-197
2	ZNF454	1.865821e-194


```

3      KBTBD7 2.417628e-193
4      STPG1 6.018686e-192
5 LOC105369537 2.551798e-191
6      LINC01351 1.679957e-189

```

```
print(paste("Found", nrow(significant_genes), "significant genes"))
```

```
[1] "Found 11248 significant genes"
```

This analysis found several genes with significant differences in expression across the three cell classes (GABAergic, Glutamatergic, Non-neuronal), using ANOVA. The top significant genes include NUP35, ZNF454, KBTBD7, STPG1, LOC105369537, and LINC01351, with p-values ranging from 2.62e-197 to 1.68e-189. These extremely low p-values indicate strong statistical evidence that these genes are expressed differently among the three cell types. Since the p-values are well below the threshold of 0.05, I can reject the null hypothesis and conclude that gene expression varies significantly across the different cell classes.

Plot for H4 (Box Plot)

```

top_genes <- significant_genes$Gene[1:5]

plot_data <- trimmed_means_data %>%
  filter(feature %in% top_genes) %>%
  pivot_longer(cols = -feature, names_to = "sample", values_to = "expression") %>%
  mutate(cell_class = case_when(
    grepl("Inh", sample) ~ "GABAergic",
    grepl("Exc", sample) ~ "Glutamatergic",
    grepl("Astro|Oligo|Micro|OPC|Endo|VLMC", sample) ~ "Non-neuronal",
    TRUE ~ "Other"
  )) %>%
  filter(cell_class != "Other")

ggplot(plot_data, aes(x = feature, y = expression, fill = cell_class)) +
  geom_boxplot() +
  labs(title = "Gene Expression Differences Across Cell Classes",
       x = "Gene",
       y = "Expression Level",
       fill = "Cell Class") +
  theme_minimal()

```



As we can see, this boxplot shows the expression levels of the top 5 significant genes (KBTBD7, LOC105369537, NUP35, STPG1, ZNF454) across the three cell classes: GABAergic (inhibitory neurons), Glutamatergic (excitatory neurons), and Non-neuronal cells. The x-axis represents the gene names, while

the y-axis shows the expression levels. The colors represent different cell classes: red for GABAergic, green for Glutamatergic, and blue for Non-neuronal cells.

A boxplot was chosen because it effectively displays the distribution of expression values for each gene across different cell classes. It shows the median, interquartile range (IQR), and potential outliers, making it easier to see differences and variability between groups.

The plot shows that most of the significant genes (like NUP35 and STPG1) exhibit noticeably higher expression levels in the Glutamatergic neurons compared to GABAergic and Non-neuronal cells. This aligns with the hypothesis that specific marker genes are associated with certain cell classes.

Plot for H4 (Density Plot)

```
top_genes <- significant_genes %>%
  head(5)

density_data <- trimmed_means_data %>%
  filter(feature %in% top_genes$Gene) %>%
  pivot_longer(cols = -feature, names_to = "sample", values_to = "expression") %>%
  mutate(cell_class = case_when(
    grepl("Inh", sample) ~ "GABAergic",
    grepl("Exc", sample) ~ "Glutamatergic",
    grepl("Astro|Oligo|Micro|OPC|Endo|VLMC", sample) ~ "Non-neuronal",
    TRUE ~ "Other"
  )) %>%
  filter(cell_class != "Other")

ggplot(density_data, aes(x = expression, fill = cell_class, color = cell_class)) +
  geom_density(alpha = 0.6) +
  facet_wrap(~feature, scales = "free") +
  labs(title = "Density Plot of Gene Expression Across Cell Classes",
       x = "Expression Level",
       y = "Density",
       fill = "Cell Class",
       color = "Cell Class") +
  theme_minimal()
```



A density plot was also chosen to visualize the data because it could show us the distribution of gene expression levels, instead of just the mean or median values. It allows us to see how gene expression varies within each cell class, making it useful for identifying whether the expression is uniform or variable within a group.

This density plot shows the expression levels of the top five significant genes (KBTBD7, LOC105369537, NUP35, STPG1, ZNF454) across three major cell classes: GABAergic (inhibitory neurons), Glutamatergic (excitatory neurons), and Non-neuronal (including cells like astrocytes, oligodendrocytes, microglia, OPCs, endothelial cells, and VLMCs).

The x-axis represents the gene expression levels, while the y-axis shows the **density** of these expression levels within each cell class. Each panel corresponds to one gene, and the color-coded curves represent the cell classes:

- Red: GABAergic (inhibitory neurons)
- Green: Glutamatergic (excitatory neurons)
- Blue: Non-neuronal (supporting cells rather than neurons)
- We can see Distinct Peaks for GABAergic Cells: Since most of the significant genes (like KBTBD7, NUP35, and ZNF454) show a high peak of density at low expression levels in GABAergic neurons, it suggests that these genes are minimally expressed in these cells.
- The density for Glutamatergic neurons is in general lower and more spread out compared to GABAergic cells, meaning there be a broader range of expression levels. Perhaps it means that the expression of these genes in excitatory neurons is more variable.
- When we look at the blue density curves, we see that non-neuronal cells typically have lower expression levels compared to neurons for these genes. This supports the idea that these genes are primarily associated with neuronal activity rather than glial or other non-neuronal functions.

Essentially, the density plot highlights that there are some significant differences between cell classes, especially between GABAergic and Glutamatergic neurons, supporting the hypothesis that specific marker genes are strongly associated with particular cell types.

Discussion

The analysis of the data set with respect to the four hypotheses this project focused on revealed distinct patterns of gene expression related to neuron subtypes, donor sex, cortical layers, and major cell classes.

For Neuron Subtypes (SST vs. VIP): I found significant differences in gene expression between SST and VIP neurons, particularly when it comes to genes like HSDL2 and RIPPLY2. This therefore points out that specific genes may indeed play unique roles in these neuron subtypes.

Donor Sex Differences: The analysis here revealed several genes, including MAGI2 and TAB1, which do show sex-related differences. These findings indicate that gene expression might be influenced by biological sex, which could have big implications for when it comes to understanding sex-specific neural functions.

Cortical Layer Variability: Genes involved in neurotransmission, like GRIN1 and SLC17A7, did show variability across different cortical layers, highlighting some potential functional differences in signal transmission depending on the cortical region.

Marker Genes and Cell Classes: We identified specific marker genes, such as NUP35 and STPG1, that showed significantly different expression levels among GABAergic, Glutamatergic, and Non-neuronal cell classes, throwing some support behind the idea that gene expression is closely linked to cell type.

Overall, all four hypotheses within this project had their null hypothesis rejected, indicating that their alternate counterparts (aka the hypothesis themselves) might have some kernel of truth to them and that those need to be further researched and analyzed to be more definitive.

References

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4. Miller, D. J., Bhaduri, A., Sestan, N., & Kriegstein, A. (2019). Shared and derived features of cellular diversity in the human cerebral cortex. *Current Opinion in Neurobiology*, 56, 117–124. <https://www.sciencedirect.com/science/article/abs/pii/S0959438818301958>
5. Source for data sets : <https://portal.brain-map.org/atlas-and-data/rnaseq/human-m1-10x>

Note:

Originally, this project was supposed to use the data matrix file that had all of the actual raw data, however, while I was able to read that file in a few times, it would stop working the next time I opened it and only worked if I re-downloaded the file from the web. This cycle continued and unfortunately I had to pivot. I chose trimmed means, because the data a bit more cleaned up and while I think the matrix could have provided better results and more ways of analysis, trimmed means was the next best thing.